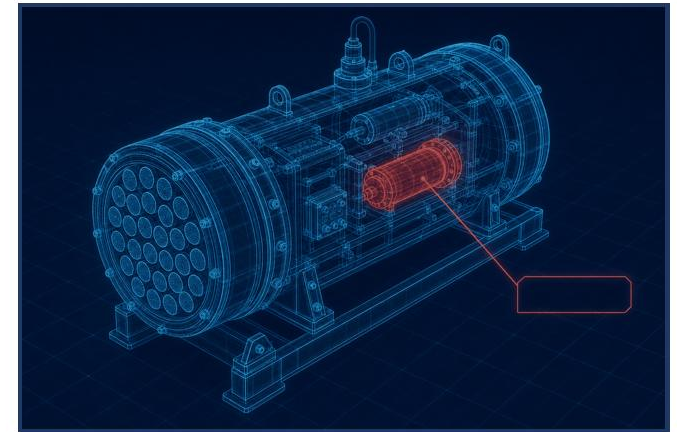


## TITLE PAGE

# SMART INDIA HACKATHON 2026

|                                  |                                                                               |
|----------------------------------|-------------------------------------------------------------------------------|
| Problem Statement ID             | 26075                                                                         |
| Problem Statement Title          | CAPACITY CONNECT — A Digital Capacity Building and Learning Management Portal |
| Theme                            | Smart Education                                                               |
| PS Category                      | Software                                                                      |
| Team ID                          | Team ID                                                                       |
| Team Name (Registered on portal) | Team Name                                                                     |

One portal for training, assessment and competency — that keeps working when the network does not.



Digital Twin concept (illustrative)

**ORGANISATION**

**Ministry of Earth Sciences  
(MoES)**

**DEPARTMENT**

**India Meteorological  
Department**

## CAPACITY CONNECT — one offline-first portal for learning, assessment, competency mapping and AI-assisted field troubleshooting

### ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

#### Detailed explanation of the proposed solution

- **Single portal, three roles. Trainee, Trainer and Admin** — secure signup/login with Admin approval and role management.
- **Trainee: professional profile (qualifications, experience, skills, interests, certificates), course enrollment, learning resources, subject-wise MCQ assessments and course feedback.**
- **Trainer: competency declaration, questionnaires with deadlines, participation and performance monitoring, and a Trainer Library of recorded lectures, presentations and study material.**
- **Admin: user approval, role management, dashboards for courses / enrollments / certifications / assessments / participation, plus homepage publishing of notifications, announcements and achievements.**
- **Competency mapping engine scores trainers against subject requirements to objectively identify the right trainer.**
- **Offline-first edge app: a real local database plus a local AI node on 127.0.0.1, so work continues at sea and at remote sites.**

#### How it addresses the problem

- Replaces scattered drives, mail and spreadsheets with one auditable source of truth for learning and competency.
- Removes the connectivity barrier: every feature works offline and reconciles automatically on reconnect.
- Turns tacit instrument know-how into retrievable, guided diagnostic procedures.



Concept UI mockup — offline dashboard and mobile learning app

#### Innovation and uniqueness of the solution

**Offline is the default, not a fallback** — the local DB is an operating database, not a cache

**Local AI engine, no cloud LLM** — deterministic retrieval pipeline on 127.0.0.1

**Learning fused with operations** — a fault maps to the asset and to the training that covers it

**3D Digital Twin guidance** — SONAR-001 → Mesh\_042 shown CRITICAL from a cached GLB

**Objective competency mapping** — fit computed from skills, certificates and outcomes

**Auditable sync ledger** — every offline action is versioned and conflict-checked



## Technologies to be used (programming languages, frameworks, hardware)

### CLIENT

- Flutter 3 / Dart, Material 3
- Riverpod state - GoRouter
- Drift + SQLite (SQLCipher)
- Android - Windows - Linux - Web

### LOCAL AI

- Python 3.11 - FastAPI on 127.0.0.1
- Normalise - detect - tokenize
- Keyword / phrase / RapidFuzz
- TF-IDF retrieval, pluggable LLM

### 3D & MEDIA

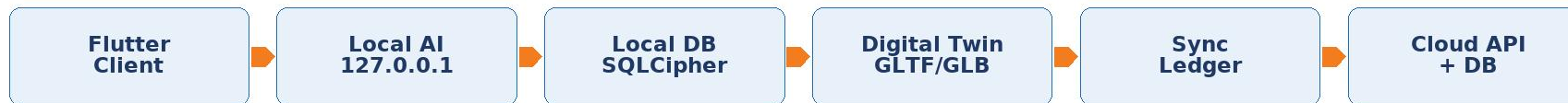
- GLTF / GLB cached locally
- Mesh mapping + fault state
- Resumable chunked downloads
- Streaming video - PDF viewer

### BACKEND & OPS

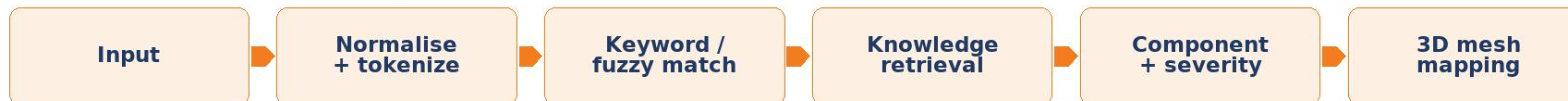
- FastAPI - SQLAlchemy - Pydantic
- PostgreSQL + Alembic
- JWT + refresh - Docker - CI
- pytest - flutter test/analyze

## Methodology and process for implementation

### A. System architecture - the edge application works with zero connectivity



### B. Local AI troubleshooting pipeline - returns structured JSON



Offline-first edge + cloud, joined by a sync ledger

Worked example: "Sonar transducer showing abnormal vibration and casing fracture" -> SONAR-001 - Mesh\_042 - severity HIGH - confidence 0.94 -> twin highlight, diagnostic procedure and record saved offline.

## Analysis of the feasibility of the idea

### Technically proven stack

Flutter, FastAPI, PostgreSQL - one Dart codebase covers Android, Windows, Linux, Web.

### Zero licensing cost

Entirely open-source - scaling costs only storage and compute, no per-seat fees.

### No exotic hardware

Runs on existing field tablets and vessel workstations; backend is containerised.

### Incremental rollout

Portal usable from day one; AI and Digital Twin extend the same schema later.



Target user: MoES / IMD field and shipboard personnel

## Potential challenges and risks -> Strategies for overcoming these challenges

| CHALLENGE / RISK                                  | MITIGATION STRATEGY                                                                                  |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Poor / no connectivity at sea and remote sites    | Offline-first architecture: local operating DB + local AI node; the sync ledger drains on reconnect. |
| Large 3D models on low-end field devices          | Decimated LOD meshes by default, full-resolution GLB optional; degrades to metadata-only view.       |
| Offline credential caching weakens authentication | Admin-approved devices only, Argon2id verifier, bounded grace window, audit + re-verification.       |
| Sync conflicts on shared records                  | Per-entity strategies with a manual-resolution queue; diagnostics are never silently overwritten.    |
| Digital literacy and user adoption                | Role-tailored simple UI, localisation-ready strings, optional voice input, in-app guided scenario.   |
| Storage exhaustion on field devices               | Storage manager with per-category usage, resumable downloads, permissioned asset removal.            |

## Potential impact on the target audience

### TRAINEES / FIELD ENGINEERS

- Learn, test and certify with zero connectivity
- Guided AI diagnostics instead of waiting for shore support
- Portable, verifiable competency profile

### TRAINERS

- One versioned library for lectures and material
- Questionnaires with deadlines and auto-scoring
- Live participation and performance view

### ADMINISTRATION / MoES

- Dashboards: enrollment, certification, assessment
- Objective trainer selection via competency mapping
- Complete audit trail for governance



Trainers and trainees on one governed platform

## Benefits of the solution (social, economic, environmental)

### SOCIAL

Equitable access to training for remote and shipboard staff; skill recognition through verifiable certificates; localisation-ready for Indian languages; safer field work through guided procedures.

### ECONOMIC

Lower instrument downtime and fewer avoidable failures; reduced travel and classroom cost; open-source stack with no licensing fees; reusable across INCOIS, NCPOR and NIOT.

### ENVIRONMENTAL

Fewer service trips and paper-free assessments; longer instrument life through timely preventive maintenance; better-maintained sensors mean higher-quality earth-science data.

**100%**

core features usable offline

**40%**

target cut in fault-resolution time

**3**

roles, one governed platform

**4**

platforms, one codebase



## Details / Links of the reference and research work

### Problem & organisational context

- SIH 2026 Problem Statement 26075 - [sih.gov.in](https://sih.gov.in)
- Ministry of Earth Sciences - [moes.gov.in](https://moes.gov.in)
- India Meteorological Department - [mausam.imd.gov.in](https://mausam.imd.gov.in)
- MoES Annual Report - capacity building programmes
- **National Education Policy 2020 - competency-based learning**

### Technology documentation

- Flutter & Dart - [docs.flutter.dev](https://docs.flutter.dev)
- Riverpod, GoRouter, Drift ([drift.simonbinder.eu](https://drift.simonbinder.eu))
- FastAPI, SQLAlchemy, Alembic, Pydantic docs
- SQLCipher - encrypted SQLite ([zetetic.net](https://zetetic.net))
- Khronos glTF 2.0 specification; PostgreSQL 16

### Standards & frameworks referenced

- SCORM / xAPI - learning-record interoperability
- IEEE 1484 LOM - learning-object metadata
- NIST SP 800-63B - identity and authentication
- NIST SP 800-38D - AES-256-GCM encryption
- OWASP ASVS + Mobile Top 10; ISO/IEC 27001 Annex A

### Domain & prior art studied

- Digital Twin for marine instrumentation (ISO 23247)
- Argo programme documentation - [argo.ucsd.edu](https://argo.ucsd.edu)
- Offline-first patterns; CRDT vs ledger-based sync
- Open-source LMS study (Moodle, Open edX)
- RapidFuzz / TF-IDF retrieval - confidence scoring



Standards, manuals and prior art reviewed

**Note:** all cited standards and libraries are open or publicly available; the solution uses no proprietary or licence-restricted component.